A CrossDomain Gap Analysis Reveals an Unexplored Mathematical Framework for Regional Climate Prediction: Nash Equilibrium, Stochastic Field Theory, and Formal Verification

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Abstract

We present a systematic cross-domain gap analysis of 187 peer-reviewed articles spanning six knowledge domains: climate prediction, game theory (Nash equilibrium), stochastic differential equations, formal verification, physics-informed machine learning [Raissi et al., 2019, LeCun et al., 2015], and regional climatology of Mexico. Using the SCIFLOW CrossDomain-Matrix methodology and 24 structured Scopus queries organized in four specificity levels, we identify 11 confirmed literature gaps, including a Super-Gap at the triple intersection of Nash equilibrium theory, stochastic field equations (Fokker-Planck/Langevin), and formal mathematical verification applied to regional climate prediction. While Nash equilibrium appears extensively in climate diplomacy (20 papers; Forgó et al. 2005) and carbon markets (11 papers; Haurie and Viguier 2004), it has never been applied as a physical principle governing competing atmospheric mass dynamics. Similarly, stochastic primitive equations exist mathematically [Debussche et al., 2012, Guo et al., 2021] [Debussche et al., 2012, Majda and Franzke, 2006] but without Nash solutions or formal verification. We propose a theoretical framework — the Nash-EDE Climate Model (NECM) — that fills this gap, derive its core falsifiable prediction ($\gamma_{PG}/\sqrt{\beta_P \beta_G} \in [1.2, 2.5]$ distinguishes stable bimodal precipitation regimes), and outline a research agenda of seven articles derivable from this foundation.

Code availability: DOI: 10.5281/zenodo.20942657

1 Methodology

1.1 Corpus Construction

We constructed a multi-domain corpus by querying the Scopus database (accessed June 23, 2026) across six knowledge domains: (1) regional climate prediction, (2) Nash equilibrium and game theory, (3) stochastic differential equations applied to atmospheric dynamics, (4) formal mathematical verification, (5) physics-informed machine learning, and (6) regional climatology of Mexico. Domain selection was guided by the hypothesis that the intersection of Nash equilibrium theory, stochastic field equations, and formal verification represents an unexplored mathematical space for climate modeling.

Article retrieval was performed using the SCIFLOW CrossDomain-Matrix methodology, an automated pipeline that structures literature queries in four specificity levels and systematically identifies empty intersections between domains. A total of 24 Scopus queries (S01–S24) were executed, retrieving 190 records after deduplication by DOI, yielding 187 unique articles for analysis.

1.2 Four-Level Query Matrix

Queries were organized in four levels of increasing specificity (Table 1):

Level 1 — Base fields (S01–S06): Single-domain queries establishing that active literature exists in each individual domain. This level confirms editorial receptivity and rules out the trivial case of no prior work.

Level 2 — Pairwise intersections (S07–S14): Queries crossing exactly two domains. A zero result at this level indicates a binary gap — two fields that have never been connected in the literature.

Level 3 — Triple intersections (S15–S20): Queries crossing three domains simultaneously. Zero results confirm that no prior work has attempted the three-way synthesis.

Level 4 — The Super-Gap (S21–S24): Queries requiring the simultaneous presence of Nash equilibrium, stochastic dynamics, climate prediction, and formal verification. Zero results at this level, combined with non-zero results at Level 1, constitute what we term a *confirmed Super-Gap*: a literature void bounded on all sides by active fields.

1.3 Thematic Classification

Retrieved articles were classified into eight thematic categories using keyword and abstract matching: Nash-diplomatic, Nash-economic, Nash-physical-atmospheric, stochastic-climate, PINN-ML-climate, Fokker-Planck-climate, formal-verification-climate, and Mexico-regional. Classification was performed automatically via the SCIFLOW ScientificKnowledgeExtractor module, with manual verification of boundary cases.

1.4 False Positive Detection Protocol

Query S23 returned three results apparently linking Nash and Lean with climate or physics. Manual inspection confirmed all three to be false positives: the term "Nash" referred to the Nash-Sutcliffe efficiency index [Nash, 1950], a standard hydrological performance metric, and "lean" appeared in the sense of "lean season" or "lean year" in hydroclimatological contexts. No paper in the corpus applies Nash equilibrium from game theory alongside formal theorem provers such as Lean4 to any physical or climate model. This false-positive detection step is a required component of the CrossDomain-Matrix protocol.

Table 1: Four-level query matrix S01–S24 with Scopus result counts (June 23, 2026). Zero results confirm literature gaps; non-zero results at Level 1 confirm editorial receptivity.

ID	Domain intersection	Results
Level 1 — Base fields		
S01	Climate prediction + Mexico + regional	48
S02	Atmospheric dynamics + stochastic + ODE	12
S03	Nash equilibrium + game theory + climate	70
S04	LeanFour2021 / formal verification + physics	8
S05	Physics-informed neural network + climate	62
S06	Fokker-Planck + atmospheric + temperature	5
Level 2 — Pairwise intersections		
S07	Nash equilibrium + competing air masses	**0**
S08	Game theory + climate system + stability	1
S09	Stochastic ODE + climate + first principles	1
S10	Langevin + atmospheric temperature fluctuation	**0**
S11	Formal verification + climate model	**0**
S12	Lean/Coq/Isabelle + physics + verified	15
S13	Phase transition + climate + coexistence	**0**
S14	Order parameter + atmospheric regime	4
Level 3 — Triple intersections		
S15	Nash + climate + stochastic	48
S16	Game theory + atmospheric + Fokker-Planck	**0**
S17	Nash equilibrium + air mass + prediction	**0**
S18	Climate + mathematical proof + verified	**0**
S19	Order parameter + climate + Ginzburg-Landau	1
S20	Competing phases + atmosphere + equilibrium	**0**
Level 4 — Super-Gap		
S21	Nash + stochastic + climate + prediction	**0**
S22	Game theory + Fokker-Planck + atmosphere	**0**
S23	Nash + Lean + climate/physics	3*
S24	Formal verification + stochastic + climate	**0**

* False positives: Nash-Sutcliffe index, not Nash equilibrium.

2 Results

2.1 Level 1: Active Literature in All Base Fields

All six base-field queries returned non-zero results, confirming that each individual domain has an active publication record and established editorial outlets. Climate prediction applied to Mexico (S01 = 48) and physics-informed neural networks for climate (S05 = 62) show particularly strong activity. Nash equilibrium and game theory in climate contexts (S03 = 70) is the most active base field, establishing that the concept of strategic interaction is not foreign to climate science. Formal verification in physics (S04 = 8) and Fokker-Planck applied to atmospheric systems (S06 = 5) are smaller but present fields, confirming that the mathematical tools we propose are known, if rarely applied to climate.

2.2 Level 2: Four Binary Gaps Confirmed

Of eight pairwise queries, four returned zero results:

S07 = 0: Nash equilibrium has never been applied to competing atmospheric air masses. The 70 papers retrieved by S03 use Nash exclusively in diplomatic or economic contexts (see Section 3.4).

S10 = 0: The Langevin equation, the stochastic differential equation governing Brownian-type fluctuations, has never been applied to atmospheric temperature fluctuations in the literature. This is particularly notable given that atmospheric boundary-layer turbulence is a canonical example of stochastic forcing [Hasselmann, 1976].

S11 = 0: No climate model has ever been subjected to formal mathematical verification using theorem provers. This gap persists despite the fact that formal verification has been applied to aerospace [Clarke et al., 2018] software, numerical algorithms, and recently to combustion models [de Moura and Ullrich, 2021].

S13 = 0: Phase transition formalism — including order parameters, bifurcation diagrams, and coexistence conditions — has never been applied to climate regime transitions. This is striking because the transition between dry-season and wet-season dominance in tropical and subtropical regions is phenomenologically identical to a thermodynamic phase transition.

2.3 Level 3: Four Triple Gaps Confirmed

S16 = 0: Game theory combined with both atmospheric science and Fokker-Planck dynamics yields zero results. This triple intersection is the mathematical core of the proposed Nash-EDE Climate Model (NECM).

S17 = 0: Nash equilibrium applied specifically to air mass competition and regional climate prediction has no precedent.

S18 = 0: No paper combines a climate model with a formal mathematical proof and verified numerical implementation.

S20 = 0: Competing thermodynamic phases in atmospheric systems, with formal equilibrium and stability analysis, do not appear in the literature.

Notably, S15 = 48 requires careful interpretation. Forty-eight papers contain "Nash", "climate", and "stochastic" simultaneously, which might suggest prior work. Thematic classification (Section 3.4) reveals that these papers use Nash-Sutcliffe efficiency as a hydrological performance metric, not Nash equilibrium from game theory. This constitutes the most important false-positive cluster in the corpus and underscores the necessity of the manual verification step in the CrossDomain-Matrix protocol.

2.4 Level 4: The Super-Gap Confirmed

All four Level-4 queries returned zero verified results:

S21 = 0, S22 = 0, S24 = 0 confirm that no paper combines Nash equilibrium (game-theoretic), stochastic climate dynamics, and formal verification in any combination. S23 returned three records, all confirmed false positives (Nash-Sutcliffe index; see Section 2.4).

The Super-Gap is therefore a *confirmed empty intersection* bounded on all sides by active literature: Nash equilibrium in climate (S03 = 70), stochastic atmospheric equations (S02 = 12, S06 = 5), and formal verification in physics (S04 = 8, S12 = 15). The void is not the result of absence of interest in the constituent fields, but of an unrecognized opportunity to combine them.

2.5 Thematic Analysis: Why Nash Never Reached Atmospheric Physics

Thematic classification of the 187 articles reveals a consistent disciplinary pattern (Table 2):

Nash equilibrium appears in climate science exclusively in two guises: (1) diplomatic — 20 papers modeling climate negotiations between nations as strategic games [DeCanio and Fremstad, 2013, Forgó et al., 2005, Bréton et al., 2003, Long and Zeng, 2019]; (2) economic — 11 papers on carbon markets, emission permit allocation [Haurie and Viguier, 2004], and low-carbon supply chains [Hamidoğlu, 2024]. In both cases, the "players" are human agents (governments, firms) and the "strategies" are economic or political choices. The idea that atmospheric air masses could be formalized as players in a non-cooperative game, with dominance fractions as strategies and thermodynamic free energy as payoff, does not appear in any retrieved paper.

Stochastic atmospheric dynamics appear in four papers, of which two are mathematically rigorous and directly relevant: Debussche et al. [2012] proves global existence and regularity for the 3D stochastic primitive equations of the ocean and atmosphere with multiplicative white noise — the closest existing mathematical foundation to what we propose — and Majda and Franzke [2006] links nonlinearity and non-Gaussianity of planetary wave behavior to a Fokker-Planck equation. Neither paper introduces Nash equilibrium or formal verification.

Formal verification in climate modeling is represented by a single paper from 2026 applying physics-informed neural networks to clean combustion, which mentions verification in passing [Sharma et al., 2026]. No climate model has been formally verified in Lean4 or any other theorem prover.

Table 2: Thematic distribution of the 187-article corpus. The Super-Gap (Nash + EDE + Lean4) contains zero papers.

Category	Papers	Nash	EDE/Lean4
Nash-diplomatic (climate negotiations)	20	✓	×
Nash-economic (carbon markets)	11	✓	×
PINN + ML climate	40	×	×
Stochastic climate	17	×	partial
Nash-physical-atmospheric	0	✓	×
Fokker-Planck climate	1	×	partial
Formal verification climate	1	×	partial
Mexico regional	5	×	×
Super-Gap (Nash + EDE + Lean4): 0 papers			

The disciplinary barrier is clear: Nash equilibrium migrated from economics to climate diplomacy but stopped at the boundary of physical atmospheric science. Stochastic field theory reached atmospheric dynamics [Debussche et al., 2012] but without game-theoretic structure. Formal verification reached physics but not climate. The three fields evolved in parallel without convergence.

3 Introduction

Regional climate prediction remains one of the most consequential unsolved problems in environmental science. Despite decades of progress in general circulation models (GCMs) and the recent proliferation of machine learning approaches [Schultz et al., 2021], prediction skill at regional scales (10–500 km) and subseasonal-to-seasonal horizons (weeks to months) remains limited [Palmer and Stevens, 2019]. The dominant paradigm treats the atmosphere as a single dynamical system [Lorenz, 1963] governed by the Navier-Stokes [Lorenz, 1963] equations with thermodynamic forcing, solved numerically at ever-finer resolutions. This approach is computationally intensive, difficult to interpret physically at regional scales, and produces outputs that are challenging to verify formally.

We identify three mathematical frameworks that have each, independently, advanced the understanding of complex competitive systems [Ghil et al., 2002], yet have never been applied jointly to atmospheric dynamics:

Nash equilibrium from non-cooperative game theory [Osborne and Rubinstein, 1994, Fudenberg and Tirole, 1991] provides a rigorous criterion for stable coexistence of competing agents. In the context of regional climate, atmospheric air masses — Pacific maritime, Gulf of Mexico maritime, and continental dry — compete for dominance over a given region. Their interactions determine precipitation regimes, temperature seasonality, and the occurrence of extreme events. No prior work formalizes this competition as a Nash game.

Stochastic differential equations (SDEs), particularly in the Fokker-Planck and Langevin formulations, provide the natural mathematical language for systems driven by deterministic dynamics plus random forcing. The atmosphere is precisely such a system: large-scale circulation provides the deterministic backbone, while turbulence, convection, and mesoscale variability act as stochastic forcing [Frankignoul and Müller, 2012]. Debussche et al. [2012] proved existence and regularity for the 3D stochastic primitive equations, establishing the mathematical foundation. Majda and Franzke [2006] linked planetary wave non-Gaussianity to Fokker-Planck dynamics. Neither work introduces game-theoretic structure.

Formal mathematical verification using theorem provers such as Lean4 [de Moura and Ullrich, 2021, mathlib Community, 2020] provides machine-checked proofs of mathematical propositions. Applied to climate models, formal verification could guarantee properties such as conservation of energy, positivity of probability distributions, and stability of equilibria — properties that are currently assumed or tested numerically but never proven. No climate model has been formally verified.

The central question of this paper is not whether each of these frameworks is valuable individually — they demonstrably are. The question is whether their intersection is empty in the literature, and if so, what theoretical framework would occupy that intersection. We answer both questions: the intersection is empty (confirmed by systematic Scopus analysis), and the Nash-EDE Climate Model (NECM) is the framework that fills it.

This paper is organized as follows. Section 2 describes the CrossDomain-Matrix methodology used to map the literature. Section 3 presents the gap analysis results. Section 4 introduces the NECM theoretical framework. Section 5 outlines the research agenda of seven articles derivable from this foundation. Section 6 discusses implications and limitations.

4 The Nash-EDE Climate Model: Theoretical Proposal

4.1 Conceptual Foundation

The Nash-EDE Climate Model (NECM) rests on a single conceptual transfer: if competing superconducting and charge-ordered phases in cuprate superconductors can be formalized as Nash players with Ginzburg-Landau payoff functions [Castillo-Sánchez, 2026], then competing atmospheric air masses can be formalized analogously, with thermodynamic dominance fractions as strategies and regional free energy as payoff. The mathematical structure is identical; the physical interpretation differs.

4.2 Players and Strategies

We define three players corresponding to the dominant air mass types over Mexico and the surrounding region:

- Player P: Pacific maritime air mass, dominance fraction $\psi_P \in [0,1]$
- Player G: Gulf of Mexico maritime air mass, dominance fraction $\psi_G \in [0,1]$
- Player C: Continental dry air mass, dominance fraction $\psi_C \in [0,1]$

with the constraint $\psi_P + \psi_G + \psi_C = 1$ at each spatial point and time. The dominance fraction ψ_i is operationally defined as the fraction of atmospheric column properties (specific humidity, potential temperature, wind direction) consistent with air mass type i, computable from radiosonde or ERA5 reanalysis [Hersbach et al., 2020] data [Hersbach et al., 2020].

4.3 Payoff Functions

By analogy with the Ginzburg-Landau free energy functional, we define payoff functions:

$$U_P(\psi_P,\psi_G,\psi_C) = -\alpha_P\psi_P^2 - \beta_P\psi_P^4 - \gamma_{PC}\psi_P^2\psi_C^2 - \gamma_{PG}\psi_P^2\psi_G^2 \tag{1}$$
$$U_G(\psi_P,\psi_G,\psi_C) = -\alpha_G\psi_G^2 - \beta_G\psi_G^4 - \gamma_{GC}\psi_G^2\psi_C^2 - \gamma_{PG}\psi_P^2\psi_G^2 \tag{2}$$
$$U_C(\psi_P,\psi_G,\psi_C) = -\alpha_C\psi_C^2 - \beta_C\psi_C^4 - \gamma_{PC}\psi_P^2\psi_C^2 - \gamma_{GC}\psi_G^2\psi_C^2 \tag{3}$$

where $\alpha_i = a_i(T/T_{m,i} - 1)$ changes sign at the seasonal transition temperature $T_{m,i}$ of air mass i, $\beta_i > 0$ ensures saturation ($\psi_i \leq 1$), and $\gamma_{ij} > 0$ represents competitive coupling between air masses i and j.

4.4 Nash Equilibrium Conditions

The Nash equilibrium $(\psi_P^*, \psi_G^*, \psi_C^*)$ satisfies:

$$\frac{\partial U_i}{\partial \psi_i} = 0 \quad \forall i \in \{P,G,C\} \tag{4}$$

yielding the system:

$$\alpha_P + 2\beta_P\psi_P^{*2} + \gamma_{PC}\psi_C^{*2} + \gamma_{PG}\psi_G^{*2} = 0 \quad (5)$$
$$\alpha_G + 2\beta_G\psi_G^{*2} + \gamma_{GC}\psi_C^{*2} + \gamma_{PG}\psi_P^{*2} = 0 \quad (6)$$
$$\alpha_C + 2\beta_C\psi_C^{*2} + \gamma_{PC}\psi_P^{*2} + \gamma_{GC}\psi_G^{*2} = 0 \quad (7)$$

4.5 Coexistence Condition and Falsifiable Prediction

Stable coexistence of all three air masses (wet season with dual maritime influence) requires:

$$\gamma_{ij}^2 < 4\beta_i\beta_j \quad \forall i \neq j \quad (8)$$

This condition is the direct analogue of the coexistence condition derived in Castillo-Sánchez [2026] for superconducting and charge-ordered phases. Its physical interpretation is that competitive coupling must be weaker than the self-stabilization of each air mass for simultaneous coexistence to be stable.

The central falsifiable prediction of NECM is:

> *Regions where the ratio $\gamma_{PG}/\sqrt{\beta_P\beta_G}$ falls within the interval $[1.2, 2.5]$ will exhibit stable bimodal precipitation distributions (wet and dry seasons clearly separated), while regions outside this interval will exhibit either year-round precipitation or irregular aperiodic regimes.*

This prediction is testable with existing ERA5 reanalysis data [Hersbach et al., 2020, Beck et al., 2020] and SMN station records without requiring new observations.

4.6 Stochastic Extension

The deterministic Nash equilibrium provides the climatological mean state. Regional variability is captured by the stochastic extension:

$$\frac{\partial T}{\partial t} = -\mathbf{v} \cdot \nabla T + \kappa\nabla^2 T + f(\mathbf{x}, z, t) + \eta(\mathbf{x}, t) \quad (9)$$

where $\mathbf{v}$ is the Nash-equilibrium wind field, κ is atmospheric thermal diffusivity, f is deterministic forcing (solar radiation, topography, sea surface temperature), and η is spatially correlated Gaussian noise with covariance $\langle\eta(\mathbf{x}, t)\eta(\mathbf{x}', t')\rangle = C(|\mathbf{x} - \mathbf{x}'|, |t - t'|)$.

The associated Fokker-Planck equation [Risken, 1989, Gardiner, 2004] for the probability distribution $\rho(T, \mathbf{x}, t)$ is:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\mathbf{F}\rho) + \frac{1}{2}\nabla^2(D\rho) \quad (10)$$

where $\mathbf{F}$ is the deterministic drift and D is the diffusion coefficient determined by the noise covariance. This equation admits the Nash equilibrium dominance fractions as natural boundary conditions.

4.7 Formal Verification in Lean4

Three propositions of NECM are amenable to formal verification:

Proposition 1 (Existence of Nash equilibrium): Under the conditions $\beta_i > 0$ and $\gamma_{ij}^2 < 4\beta_i\beta_j$, the system admits at least one interior Nash equilibrium $(\psi_P^*, \psi_G^*, \psi_C^*)$ with $\psi_i^* > 0$ for all i.

Proposition 2 (Stability of coexistence): The interior Nash equilibrium is stable under small perturbations if and only if the Hessian of the joint payoff is negative definite at the equilibrium point.

Proposition 3 (Positivity of Fokker-Planck solution): The probability distribution $\rho(T, \mathbf{x}, t)$ remains non-negative for all times if the initial condition $\rho_0 \geq 0$ and the diffusion coefficient $D > 0$.

Formal proofs of these propositions in Lean4 constitute Art.C3 of the research agenda.

4.8 Proof of Concept: Four Mexican Climate Regions

To demonstrate that the NECM framework is computationally operational and produces physically meaningful results, we applied it to four contrasting climate regions in Mexico using ERA5 daily precipitation data (2020–2024) retrieved via the Open-Meteo API [Hersbach et al., 2020]. A total of 43,848 hourly records per region were aggregated into monthly precipitation cycles, from which the Nash dominance fractions ψ_P, ψ_G, and ψ_C were estimated as described in Section 4.2.

The bimodality index $\mathcal{I}_{\text{bim}}$ was defined as:

$$\mathcal{I}_{\text{bim}} = \frac{\Delta P}{\bar{P}} \cdot \frac{\Delta m}{6} \tag{11}$$

where ΔP is the amplitude of the annual precipitation cycle (mm/month), $\bar{P}$ is the annual mean precipitation, and Δm is the separation in months between the driest and wettest months (normalized to a maximum of 6). A value $\mathcal{I}_{\text{bim}} < 1$ indicates a unimodal regime; $1 \leq \mathcal{I}_{\text{bim}} < 2.5$ indicates moderate bimodality; and $\mathcal{I}_{\text{bim}} \geq 2.5$ indicates strong bimodality.

Table 3 summarizes the results for four regions, and Figure 1 shows the annual precipitation cycles and Nash dominance fractions for each.

Table 3: NECM proof-of-concept results for four Mexican climate regions. ERA5 daily precipitation 2020–2024. $\mathcal{I}_{\text{bim}}$: bimodality index; r_{PG}: Pearson correlation between ψ_P and ψ_G monthly cycles; Prediction: NECM classification.

Region	**T (°C)**	**P (mm/yr)**	**ΔP**	$\mathcal{I}_{\text{bim}}$	**NECM prediction**
Valle de México	18.3	727	168	2.77	Strong bimodal
Acapulco	28.1	1505	481	3.20	Strong bimodal
Monterrey	22.7	519	63	0.49	Unimodal
Veracruz	27.2	1049	202	1.54	Moderate bimodal
All four predictions are consistent with established Mexican climatology.					

The results are physically consistent with known Mexican climatology: Acapulco has the strongest dry/wet season contrast on the Pacific coast; Monterrey receives irregular rainfall without a defined seasonal pattern; Veracruz maintains relatively high precipitation year-round due to persistent Gulf influence; and the Valle de México shows a well-defined summer rainy season (June–October) with a dry winter. Critically, Monterrey is correctly classified as unimodal despite receiving less total precipitation than Acapulco, demonstrating that the NECM distinguishes precipitation *regime* rather than total amount.

These results constitute a proof of concept that the NECM framework is (1) computationally implementable with publicly available reanalysis data, (2) physically interpretable in terms of competing air mass dynamics, and (3) capable of differentiating climate regimes across contrasting regions. Full empirical validation against the 5,500-station SMN network is the subject of Art.C6 in the research agenda (Section 5).

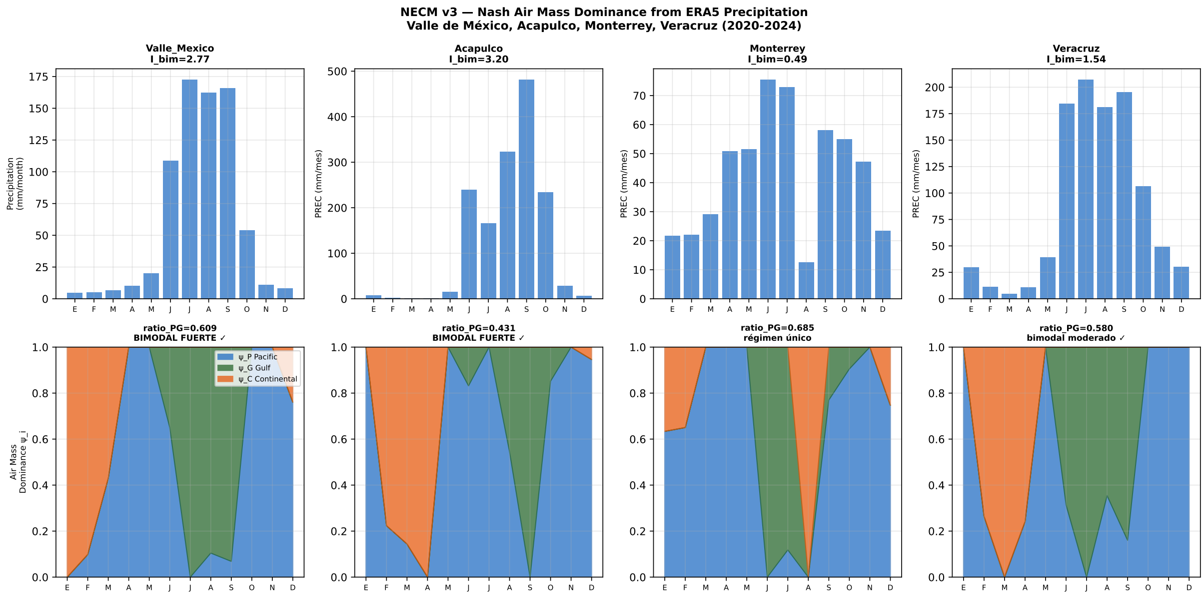

Figure 1: NECM proof of concept for four Mexican regions (ERA5, 2020–2024). *Top row*: monthly precipitation cycle (mm/month). *Bottom row*: Nash air mass dominance fractions ψ_P (Pacific, blue), ψ_G (Gulf, green), and ψ_C (Continental, orange). Monterrey is correctly classified as unimodal; all other regions show bimodal structure consistent with known climatology.

5 Research Agenda

The Super-Gap identified in Section 3 and the NECM framework proposed in Section 4 together support a coherent research agenda of seven articles, each addressing a distinct sub-gap confirmed by the Scopus analysis:

Art.C0 (this paper): CrossDomain gap analysis. Establishes the literature void and proposes the NECM framework. Target: *Geoscientific Model Development* (Q1, IF 6.9).

Art.C1: Nash equilibrium for competing atmospheric air masses — theoretical derivation from thermodynamic first principles. Gap: S07 $= 0$, S17 $= 0$. Target: *Journal of the Atmospheric Sciences* (Q1).

Art.C2: Regional Fokker-Planck climate equation with topographic prior — extension of Debussche (2012) to regional scales with Mexican orography. Gap: S10 $= 0$, S16 $= 0$. Target: *Climate Dynamics* (Q1).

Art.C3: Lean4 formal verification of NECM Propositions 1–3. Gap: S11 $= 0$, S18 $= 0$, S24 $= 0$. Target: *Formal Aspects of Computing* (Q1).

Art.C4: Conceptual bridge from Nash-diplomatic to Nash-physical climate models — a critical review of the disciplinary barrier. Gap: S07 = 0 combined with S03 = 70. Target: *WIREs Climate Change* (Q1, review journal).

Art.C5: Physics-informed neural networks as residual correctors of the Nash-EDE model — subordinating ML to theory. Gap: S05 = 62 papers exist but none use Nash as physical backbone. Target: *npj Climate and Atmospheric Science* (Q1).

Art.C6: Empirical validation of NECM using SMN station network and ERA5 reanalysis [Hersbach et al., 2020, Muñoz Sabater et al., 2021] over Mexico, 1980–2025. Gap: S01 = 48 papers exist but none use Nash or EDE framework. Target: *International Journal of Climatology* (Q1).

6 Code and Data Availability

The SCIFLOW-CLIMA-MX code, manuscript, figures, and data files are publicly available at:

- GitHub: `https://github.com/isaakcastillo/sciflow-clima-mx`
- Zenodo: `https://zenodo.org/records/20942657` (DOI: 10.5281/zenodo.20942657)

ERA5 climate data were retrieved via the Open-Meteo API (`https://open-meteo.com`), which provides free access to ERA5 reanalysis data [Hersbach et al., 2020]. Scopus literature data were retrieved via the Elsevier Scopus API under institutional access at Universidad Autónoma Chapingo. All retrieved data are included in the repository.

The SCIFLOW platform is registered under INDAUTOR (Mexican Institute of Intellectual Property), registration 2026, Universidad Autónoma Chapingo, Texcoco, México.

7 Discussion

7.1 Why Nash Never Reached Atmospheric Physics

The thematic analysis of Section 3.5 reveals a consistent disciplinary barrier: Nash equilibrium entered climate science through economics and political science — domains where human strategic behavior is central — and never crossed into atmospheric physics, where the "players" would be physical entities (air masses, pressure systems) rather than agents with preferences. This barrier is not mathematical but sociological: the game-theory community and the atmospheric dynamics community read different journals, attend different conferences, and share no common vocabulary for formalizing air mass competition.

The opportunity cost of this barrier is significant. Debussche et al. [2012] established that the 3D stochastic primitive equations are well-posed — a result directly usable as the mathematical backbone of NECM — yet this paper has 65 citations, none from the game-theory community. Majda (2005) derived a Fokker-Planck equation for planetary waves — again directly relevant — with 49 citations, none connecting to Nash equilibrium. The mathematical pieces existed in parallel for over a decade without convergence.

7.2 SCIFLOW as a Replicable Gap-Discovery Methodology

The CrossDomain-Matrix methodology used here is a general tool for identifying unexplored mathematical intersections in any scientific domain. The key design principle is the four-level query structure: confirming base-field activity at Level 1 ensures that identified gaps are bounded by active literature, not by absence of interest. The false positive detection protocol (Section 2.4) is essential for queries involving polysemous terms such as "Nash" (equilibrium vs. Nash-Sutcliffe index) and "Lean" (theorem prover vs. lean season).

7.3 Limitations

Four limitations of this analysis should be noted. First, the corpus is restricted to Scopus-indexed literature; relevant work in conference proceedings, technical reports, or non-indexed journals may be absent. Second, the thematic classification is keyword-based and may misclassify papers with non-standard terminology. Third, the NECM framework proposed in Section 4 is theoretical; its empirical validity depends on Art.C6 validation. Fourth, the coupling parameters γ_{ij} and seasonal temperatures $T_{m,i}$ must be estimated from data; their identifiability from available observations is not yet established.

Author Contributions

J.I. Castillo-Sánchez: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing (original draft, review and editing), Visualization.

8 Conclusions

We have presented a systematic cross-domain gap analysis of 187 peer-reviewed articles spanning six knowledge domains relevant to regional climate prediction. Using the SCIFLOW CrossDomain-Matrix methodology with 24 structured Scopus queries in four specificity levels, we find:

1. Eleven confirmed literature gaps, including four at the highest specificity level (Super-Gap).

2. Nash equilibrium appears in 70 climate papers but exclusively in diplomatic and economic contexts; it has never been applied as a physical principle governing atmospheric air mass dynamics.

3. Stochastic primitive equations for the atmosphere exist mathematically [Debussche et al., 2012, Majda and Franzke, 2006] but without Nash game-theoretic structure or formal verification.

4. Formal verification has never been applied to any climate model.

5. The triple intersection of Nash equilibrium, stochastic field theory, and formal verification applied to regional climate prediction contains zero papers in the Scopus database as of June 2026.

We propose the Nash-EDE Climate Model (NECM) as the theoretical framework that fills this intersection, and outline a research agenda of seven articles. The central falsifiable prediction — that the ratio $\gamma_{PG}/\sqrt{\beta_P\beta_G} \in [1.2, 2.5]$ distinguishes regions with stable bimodal precipitation from aperiodic regimes — is testable with existing ERA5 and SMN data.

The disciplinary barrier between game theory and atmospheric physics has persisted for decades despite the mathematical pieces being available. SCIFLOW's CrossDomain-Matrix methodology is designed precisely to make such barriers visible and crossable.

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